ETSI FRMCS
Application Session Concept

Proposed Modifications to ETSI TS 103 764, TS 103 765-3/4/5

Working Contribution – Draft v2.4

R2DATO D25.3 – Concept Paper FRMCS

Target documents: ETSI TS 103 764 (RT-0095), TS 103 765-3 (RT-0097), TS 103 765-4 (RT-0099), TS 103 765-5 (RT-00104)

Unchanged documents: All UIC specifications (FFFIS-7950, FIS-7970, FRS FU-7120, AT-7800, TOBA FRS-7510)

Contents

1 Architectural Principle

1.1 Reformulation of the Session Concept

In the ETSI TSs, **"session"** previously referred to the MCData IPcon tunnel established by the Gateway on behalf of an application (what this proposal calls a **service session**). This proposal redefines the terms:

- **"session"**: the application session – communication between two FRMCS railway applications, composed of one or more service sessions managed internally by the Gateways.
- **"service session"**: an individual communication bearer. The first is necessarily MCx (MCData IPcon) – unchanged. Additional ones may be any MCx type or future railway critical services (R2DATO D25.3 Part B); not limited to the same service domain.

All internal MCx tunnel occurrences of "session" in the ETSI TSs are replaced by "service session". Application-visible identifiers (`sessionId`, `POST /sessions`, `openSession*`, `closeSession*`, `incomingSessionNotif*`, `open session`) are **unchanged**.

Nothing changes for applications. OB_{APP} and TS_{APP} interfaces are strictly unchanged. UIC specs already use "session" in the application sense – their terminology is now naturally aligned without modification.

1.2 Why ASAS Rather than SIP INVITE AnyExt

RT-0097 §6.2.2.3 already carries `<application-data>` in the `AnyExt` field (format "not specified in the present document"). While usable for MCx, this approach is structurally excluded: a future non-MCx critical service has no SIP INVITE to carry the negotiation. The protocol-agnostic ASAS is the only mechanism that survives the application/service strata separation of R2DATO D25.3 Part B.

1.3 Application Identifier Resolution (AIR)

The AIR is an internal Gateway function resolving `remoteId` to a **service identifier** (MCx or future critical service). **First-edition rules – fully specified:**

1. If `remoteId` ends with `.ertms` (ETCS format `id.ty.cc.ertms`, UIC AT-7800): append the current MCx domain ⊠ `id.ty.cc.ertms@current_domain`. Consistent with current UIC specifications; functional baseline (note: non-optimal in domain-overlap zones).
2. Otherwise: pass-through as MCx FA – identical to historical behaviour.

The AIR function is designed as an **extension point**. Future editions may introduce DNS-based or FROP-configured resolution without any change to OB_{APP}, TS_{APP}, SIP, or UIC specifications (ERJU/OCORA).

1.4 Initialisation Mechanism

Once the first service session is established (SIP + GRE unchanged), each Gateway exposes an **ASAS** reachable on its post-NAT IP address (SD_PUB_IP, cf. §2.5). The calling Gateway drives:

1. **Initialisation**: POST to ASAS with `svcSessionId_1` as correlator; receives `appSessionId_term`, Ed25519 public key, alias list, multipath policy, keepalive period.
2. **Additional service sessions**: established per alias list + FROP; each declared to ASAS with signed `sessionToken`.

3. **Keepalive**: sent only if applicative silence $\geq$ T_KA on a given `svcSessionRef` (intelligent keepalive, cf. §2.4).
4. **Alias list update**: PATCH sent immediately by terminating GW on any new alias or service identifier registration. No configuration required.
5. **Release**: DELETE to ASAS + release each service session per its protocol.

1.5 Application Session Lifecycle

A session is alive while at least one service session is `ACTIVE` or `DEGRADED`. When all are lost, a **recovery timer** starts: INTEGER(1..180) s (FROP); SHALL be ≤ 5 s for sessions carrying ETCS/ATP (AT-7800 Annex A.6).

2 Key Concepts

2.1 Terminology

Term	Definition
session	Application-level communication between two FRMCS railway applications, composed of one or more service sessions. Identified externally by `sessionId` (OB_{APP}/TS_{APP}), internally by (`appSessionId_call`, `appSessionId_term`).
service session	Individual communication bearer, transparent to applications. First is necessarily MCx (MCData IPcon). Additional may be MCx or future critical services. Identified by `svcSessionRef` within its parent session.
open session	A session that has not been released (unchanged).
AIR	Internal Gateway function resolving `remoteId` to a service identifier (MCx or future critical service). First-edition rules (fully specified): `*.ertms` ⊠ `*.ertms@current_domain`; otherwise pass-through FA MCx. Extension point (DNS/FROP, ERJU/OCORA). Not visible to applications.
ASM	Internal Gateway function managing session and service session lifecycle, driving ASAS interactions. Not visible to applications.
ASAS	REST/TLS API server on the Gateway's post-NAT IP (SD_PUB_IP). Trust anchored by first MCx service session. Ed25519 key exchange sufficient – no inter-GW PKI required. Absorbs `FUTURE_CRITICAL_SERVICE` without modification.
Calling GW	GW initiating `POST /sessions` via OB_{APP}. Derives ASAS IP from post-NAT SIP source IP.
Terminating GW	GW hosting destination application. Exposes ASAS. Builds alias list; SHALL immediately PATCH on any new alias or service identifier registration; notifies application via TS_{APP} (`incomingSessionNotif*`).
alias list	Dynamic list of service identifiers + types + domains. Returned at initialisation; immediately updated via `PATCH /aliases` on any new registration – no FROP configuration required.

Term	Definition
service identifier	Identifier used to establish a service session, regardless of service type. For MCx: FA or MC Service User ID. For future critical services: any identifier the terminating Gateway can associate with the destination user.
recovery timer	INTEGER(1..180) s (FROP); SHALL be ≤ 5 s for ETCS/ATP (AT-7800 Annex A.6). Session released on expiry.
T_KA	Keepalive period: INTEGER(1..10) s (application profile). Rule: T_KA $\leq$ recovery_timer / 2 (≤ 2.5 s for ETCS/ATP). $N = 2$ consecutive failures (fixed) ⊠ `DEGRADED`.

2.2 Session Identifiers

- `appSessionId_call` (UUID v4): calling GW, stable for session lifetime.
- `appSessionId_term` (UUID v4): terminating GW at ASAS init. Analogous to GTP TEIDs.
- `svcSessionRef` (UUID v4): identifies a service session. **Stable and immutable**, including during recovery.
- `svcSessionId` (native): SIP Call-ID for MCx. Used as correlator. May change during recovery.

2.3 Trust Model and Authentication

The first MCx service session is a trusted service already responsible for identifying the parties – its establishment is a sufficient trust anchor.

Level 1 – Channel Confidentiality (TLS 1.3)

Server certificate may be **self-signed** – no inter-operator PKI required.

> **FSNNI topology hiding note**: a potential intermediate domain performing topology hiding does not compromise trust – the Ed25519 `sessionToken` provides per-message integrity regardless of the network path. The only constraint is ASAS reachability on SD_PUB_IP (covered by §2.5).

Level 2 – Trust Anchor (First MCx Service Session)

The MC Service Server authenticated the calling Gateway before accepting the session. Ed25519 public key exchange over this channel bootstraps trust.

Level 3 – Per-Message Authentication (Ed25519 sessionToken)

```
sessionToken = Sign(sk_call,
  appSessionId_call || 0x00 || appSessionId_term || 0x00 ||
  svcSessionRef     || 0x00 || svcSessRequestTime || 0x00 ||
  svcSessAcceptTime)
// Recovery: append 0x00 || replSvcSessionRef
ackToken = Sign(sk_term, same input)
```

Serialisation: UTF-8 concatenation with `0x00` separators – deterministic, no external dependency.

2.4 Intelligent Keepalive

- T_KA: INTEGER(1..10) s (application profile); SHALL be ≤ 2.5 s for ETCS/ATP.
- $N = 2$ consecutive failures (fixed) ⊠ `DEGRADED`.
- Recovery: new service session declared with same `svcSessionRef` + signed token.

2.5 NAT Addressing Model

ASAS address rule (resolved): post-NAT source IP of first service session SIP signalling (SD_PUB_IP, RT-0097 §6.2.2.4). No SIP header extraction required. Port configurable in FROP.

3 Terminology Replacement Rule in ETSI TSs

Context in TSs	Current term	Proposed term
Internal MCx bearer	session	**service session**
Internal MCx establishment	session establishment	**service session** establishment
Internal MCx release	session release	**service session** release
Internal Multipath procedure	Multipath session	Multipath **service session**
OB_{APP}/TS_{APP} identifier	`sessionId`	unchanged
Application request	`POST /sessions`, `openSession*`	unchanged
Incoming notification	`incomingSessionNotif*`	unchanged
Application release	`closeSession*`, `DELETE /sessions`	unchanged
Application state	`open session`	unchanged

4 Proposed ETSI Modifications

4.1 ETSI TS 103 764 (RT-0095) – Normative Annex X

X.1 – Scope ASAS inter-GW API, agnostic to service session type. Not visible to applications. Does not modify any UIC specification or service protocol. Absorbs `FUTURE_CRITICAL_SERVICE` without modification of this annex.

X.2 – Transport and Trust Model **Transport:** TLS 1.3, HTTP/2 + JSON. Port: FROP.

Server certificate: Self-signed or any locally-deployed CA.

ASAS IP address: Post-NAT source IP of first service session SIP signalling (SD_PUB_IP, RT-0097 §6.2.2.4). No SIP header extraction required.

Trust model: First MCx service session anchor + Ed25519 keys. No inter-GW PKI required.

Token serialisation: UTF-8 fields concatenated with `0x00` separators (see §2.3).

Timers: recovery timer INTEGER(1..180) s (FROP); T_KA INTEGER(1..10) s (application profile); $N = 2$ (fixed).

mTLS option: If bilateral OOB FSNNI agreement (FROP).

X.3 – Initialisation (`POST /asas/appsessions`)

```
// Request
{ "appSessionId_call": "uuid-v4", "svcSessionRef_1": "uuid-v4",
  "svcSessionId_1": "native-id", "callingPublicKey": "base64(Ed25519)",
  "keepaliveReq": 2,
  "mpCapabilities": { "supportedPolicies": ["DUPLICATION","RECOVERY"], "
   maxPaths": 4 } }

// Response 200 OK
{ "appSessionId_term": "uuid-v4", "svcSessionRef_1": "uuid-v4",
  "terminatingPublicKey": "base64(Ed25519)", "keepaliveAccepted": 2,
  "mpPolicy": { "policy": "DUPLICATION|RECOVERY|LOADBALANCE|NONE", "
   activePaths": 2 },
  "advertisedAliases": [{ "svcSessionRef": "uuid-v4",
    "serviceType": "MCX|FUTURE_CRITICAL_SERVICE",
    "serviceDomain": "string", "serviceId": "string",
    "reachability": "CURRENT|ALTERNATIVE" }] }
```

X.4 – Service Session Declaration / Recovery

```
// Request
{ "appSessionId_call": "uuid-v4", "appSessionId_term": "uuid-v4",
  "svcSessionRef": "uuid-v4", "serviceType": "MCX|FUTURE_CRITICAL_SERVICE
   ",
  "svcSessionId": "native-id", "svcSessRequestTime": "ISO8601",
  "svcSessAcceptTime": "ISO8601",
  "sessionToken": "base64(Sign(sk_call, quintet_0x00_separated))" }
// Recovery: add "replSvcSessionRef": "uuid-v4"

// Response 200 OK
{ "svcSessionRef": "uuid-v4",
  "ackToken": "base64(Sign(sk_term, same input))" }
```

X.5 – Keepalive Sent only if applicative silence ≥ T_KA on listed `svcSessionRef`.

```
{ "appSessionId_call": "uuid-v4", "appSessionId_term": "uuid-v4",
  "svcSessionRefs": ["uuid-v4"], "timestamp": "ISO8601" }
// Response: { "results": [{ "svcSessionRef": "uuid-v4", "status": "
   ACTIVE|DEGRADED" }] }
```

X.6 – Alias List Update (`PATCH .../aliases`) Sent **immediately** by terminating GW on any new alias or service identifier registration during an active session. No FROP configuration required.

```
{ "appSessionId_call": "uuid-v4", "appSessionId_term": "uuid-v4",
  "updatedAliases": [{ "svcSessionRef": "uuid-v4",
    "serviceType": "MCX|FUTURE_CRITICAL_SERVICE",
    "serviceDomain": "string", "serviceId": "string",
    "reachability": "CURRENT|ALTERNATIVE", "action": "ADD|REMOVE" }] }
```

X.7 – Release (`DELETE .../appsessions/{id}`)

```
{ "appSessionId_call": "uuid-v4", "appSessionId_term": "uuid-v4",
  "reason": "NORMAL|ERROR|TIMEOUT" }
```

X.8 – State Machines Session:

```
[CREATING]    -- 200 OK            --> [ESTABLISHED]
```

```
[CREATING]     -- 4xx/5xx          --> [FAILED]
[ESTABLISHED] -- all svc lost     --> [DEGRADED]
[ESTABLISHED] -- DELETE /sessions --> [RELEASING]
[DEGRADED]     -- >=1 ACTIVE       --> [ESTABLISHED]
[DEGRADED]     -- timer expired    --> [RELEASING]
[RELEASING]    -- done             --> [RELEASED]
```

Service session:

```
[CREATING]          -- bearer+ASAS OK    --> [ACTIVE]
[ACTIVE]            -- 2 KA no response --> [DEGRADED]
[DEGRADED]          -- recovery          --> [ACTIVE]
[ACTIVE|DEGRADED] -- RELEASING         --> [RELEASED]
```

X.9 – Error Mapping to OB_{APP} Causes

ASAS cause	`openSessionFinalAnswerNotif` cause
ASAS timeout	`MCX_ENDPOINT_NOT_REACHABLE`
ASAS 403	`REMOTE_ENDPOINT_DECLINED`
ASAS 404	`TERMINATING_APPLICATION_ENDPOINT_NOT_REACHABLE`
ASAS 401	`TERMINATING_APPLICATION_ENDPOINT_NOT_ALLOWED`

5 Impact Summary

Document	Nature of modification	UIC specs
ETSI TS 103 764	Session + service session + AIR + ASAS + service identifier (§3.1) + ASM + Annex X	
ETSI TS 103 765-3	Systematic replacement + Step 1 (AIR⊠service id + SD_PUB_IP)	unchanged
ETSI TS 103 765-4	Systematic replacement + Step 0 + immediate alias list mgmt	unchanged
ETSI TS 103 765-5	Systematic replacement + Multipath binding	unchanged
UIC FFFIS-7950	No modification	✓
UIC FIS-7970	No modification	✓
UIC FRS FU-7120	No modification	✓
UIC AT-7800 SRS	No modification	✓
UIC TOBA FRS-7510	No modification	✓
3GPP TS 24.582/24.379	No modification	

6 Open Points

All open points are resolved. The document is ready for ETSI contribution.

~~1.~~ ASAS IP / NAT – Resolved (§2.5).

~~2.~~ FSNNI multi-hop trust – Resolved (§2.3).

~~3.~~ sessionToken serialisation – Resolved (§2.3 + §X.2).

~~4.~~ AIR rules – Resolved: `*.ertms` ⊠ `@current_domain`; otherwise pass-through.

~~5.~~ Recovery timer – Resolved: INTEGER(1..180) s; ≤ 5 s for ETCS/ATP.

~~6.~~ T_KA / N – Resolved: INTEGER(1..10) s; $N = 2$; T_KA $\leq$ timer/2.

~~7.~~ Alias list triggering – Resolved: immediate, no FROP.

~~8.~~ Extensibility – Resolved: `serviceType` open field.